Reg. No. | | | | | | | | | | | [17MT44]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

PRINCIPLES OF EXTRACTIVE METALLURGY-I

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions by choosing at least **TWO** questions from each part.
2. Missing data may be suitably assumed.
3. Use of graph sheets is permitted.

PART – A

1. a. Give examples of primary metals, secondary metals, light metals, refractory metals and rare-earth metals. **05**

b. With the aid of flow-sheet discuss the extraction of a metal from its ore. **09**

2. a. Discuss the matte smelting and converting for Cu production. State why blast furnace operation is not used for Cu extraction. **09**

b. Discuss types of roasting. What is the specific advantage of sulphatising roasting? **05**

3. a. Would an atmosphere containing 15% CO_2, 5% CO and 80% N_2 oxidise Ni at 1000 K? Given: **06**

$Ni\ (s) + 0.5O_2\ (g) = NiO\ (s) \rightarrow K_e = 5.76 \times 10^7$ (at 1000 K)

$CO\ (g) + 0.5O_2\ (g) = CO_2\ (g) \rightarrow K_e = 1.68 \times 10^{10}$ (at 1000 K)

b. What is predominance area diagram? Illustrate the diagram for Ni-S-O system. **08**

4. a. Define interaction co-efficient. In a binary dilute solution containing solutes i and j, show that $\varepsilon_i^j = \varepsilon_j^i$, where ε_i^j and ε_j^i refer to interaction co-efficient of j on i and interaction co-efficient of i on j, respectively. **09**

b Estimate the activity of sulphur in hot metal of a blast furnace containing 0.04%S, 4%C, 2%Si and 1% Mn at 1600°C in 1wt.% standard state. **05**

Given: $e_S^C = 0.24$, $e_S^{Si} = 0.066$, $e_S^S = -0.028$, $e_S^{Mn} = -0.025$.

PART-B

5. a. Derive the relationship between the degree of a heterogeneous reaction and reaction time if the reaction follows contracting sphere model. **08**

b. A reaction with an activation energy of 140 $kJmol^{-1}$ has the rate constant of $1.5s^{-1}$ at 700 K. Calculate its rate constant at 1200 K. **06**

6. a. What is leaching? Write short notes on; **08**

i. Pressure leaching,

ii. Microbial leaching and

iii. Pachuka tank.

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b. Define the following: **06**

i. solvent extraction,

ii. cementation and

iii. ion exchange.

7. **a.** What is decomposition voltage (V_D)? Derive V_D for electrolysis of aqueous solution of $CdCl_2$. **08**

b. Define the following: **06**

i. electro-winning,

ii. electro-refining and

iii. electro-forming.

8. **a.** Discuss briefly about the non-aqueous solvent extraction. **08**

b. Compare the solvometallurgy and hydrometallurgy unit processes. **06**